

I. គណនាលីមីតខាងក្រោម

- $\lim_{x \rightarrow 10} 2027$
- $\lim_{x \rightarrow 3} (2x^2 - 5x + 10)$
- $\lim_{x \rightarrow 2} \frac{x^2 + 5x + 6}{x - 1}$
- $\lim_{x \rightarrow 3} \frac{kx^2 + 9}{x}$
- $\lim_{x \rightarrow 2} \frac{x + 2}{x + 3}$
- $\lim_{h \rightarrow 0} (h^2 + 4hx - 2h + 2012x)$
- $\lim_{x \rightarrow 4} \sqrt[3]{\frac{x^2 - 3x + 4}{2x^2 - x - 1}}$
- $\lim_{x \rightarrow 2} \sqrt{x + \sqrt{x + \sqrt{x + 2}}}$

II. គណនាលីមីតខាងក្រោម

- $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$
- $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$
- $\lim_{x \rightarrow -2} \frac{x^2 - 4}{x + 2}$
- $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$
- $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4}$
- $\lim_{x \rightarrow -2} \frac{x^2 - 4}{x^3 + 8}$
- $\lim_{x \rightarrow 5} \frac{x^3 - 125}{x - 5}$
- $\lim_{x \rightarrow 2} \frac{x^6 - 64}{x^3 - 8}$
- $\lim_{x \rightarrow -4} \frac{x^3 + 64}{x^2 - 16}$
- $\lim_{x \rightarrow 0} \frac{(h + x)^3 - h^3}{x}$
- $\lim_{x \rightarrow a} \frac{x^3 - a^3}{3a - 3x}$
- $\lim_{x \rightarrow 0} \frac{(x - 2)^3 + 8}{x^3 + x}$
- $\lim_{x \rightarrow 1} \frac{x^2 + 2x - 3}{x - 1}$
- $\lim_{x \rightarrow -2} \frac{x^3 + 3x^2 + 2x}{x^2 - x - 6}$
- $\lim_{x \rightarrow -1} \frac{x^2 - 7x - 8}{2x^2 + 5x + 3}$
- $\lim_{x \rightarrow 2} \frac{2x^2 + 6x - 20}{3x^2 - 5x - 2}$
- $\lim_{x \rightarrow 1} \frac{x^3 + 3x^2 - 4}{x^3 - 1}$
- $\lim_{x \rightarrow 3} \frac{x^3 - 5x - 12}{x - 3}$
- $\lim_{x \rightarrow 2} \frac{x^3 - x^2 + 2x - 8}{x - 2}$
- $\lim_{x \rightarrow -2} \frac{2x^3 + 5x^2 + 3x + 2}{x^3 + 8}$
- $\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{5x^3 - 16x^2 + 4x - 3}$
- $\lim_{x \rightarrow 1} \frac{2x^3 - 7x^2 - 3x + 8}{5x^3 - 13x^2 - 7x + 15}$
- $\lim_{x \rightarrow -1} \frac{x^3 - 3x^2 - 3x + 1}{x^3 + 2x^2 - 3x - 4}$
- $\lim_{x \rightarrow 1} \frac{2x^3 + 3x^2 - 3x - 2}{x^3 - x^2 - 2x + 2}$
- $\lim_{x \rightarrow 2} \frac{2x^3 - 4x^2 + 2x - 4}{x^3 + 1}$
- $\lim_{x \rightarrow 0} \frac{(2x + 1)^3 - 1}{x}$
- $\lim_{x \rightarrow \frac{1}{2}} \frac{10x - 5}{2x^2 - \frac{1}{2}}$
- $\lim_{x \rightarrow a} \frac{x^2 - (a + 1)x + a}{x^3 - a^3}$
- $\lim_{x \rightarrow 1} \frac{x^{2014} - 1}{x^{2015} - 1}$
- $\lim_{x \rightarrow 1} \frac{x^n - 1}{x^m - 1} \quad n, m \in \mathbb{N}$
- $\lim_{x \rightarrow -1} \frac{x^5 + 1}{x^3 + 1}$
- $\lim_{x \rightarrow -1} \frac{x^7 + 1}{x^9 + 1}$

III. គណនាលីមីតខាងក្រោម

- $\lim_{x \rightarrow 1} \frac{x^{2015} + 1}{x^{2013} + 1}$
- $\lim_{x \rightarrow 1} \frac{x^{2019} + 10x - 11}{x^9 - 5x + 4}$
- $\lim_{x \rightarrow 1} \frac{x^{11} - 13x^{10} + 12}{x^4 + 7x - 8}$
- $\lim_{x \rightarrow 1} \frac{x^{16} + 2x^{10} - 3}{x^{19} - 15x + 14}$
- $\lim_{x \rightarrow 1} \frac{x^{11} + x^{10} - 2}{x^{100} + 5x^{12} - 6}$
- $\lim_{x \rightarrow 1} \frac{x + x^2 + x^3 - 3}{x + x^2 + x^3 + x^4 + x^5 - 5}$
- $\lim_{x \rightarrow 1} \frac{x + x^2 + x^3 + x^4 - 4}{x + x^2 + \dots + x^7 - 7}$
- $\lim_{x \rightarrow 1} \frac{x + x^2 + \dots + x^n - n}{x + x^2 + \dots + x^m - m}, \quad n, m \in \mathbb{N}$
- $\lim_{x \rightarrow 1} \frac{\sqrt{x+3} - 2}{1 - \sqrt{x}}$
- $\lim_{x \rightarrow -2} \frac{\sqrt{2x+20} - 4}{x^2 - 4}$
- $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$
- $\lim_{x \rightarrow 1} \frac{\sqrt{x+3} - 2}{x - 1}$
- $\lim_{x \rightarrow 2} \frac{\sqrt{x+7} - 3}{x^2 - 6 + x}$
- $\lim_{x \rightarrow 2} \frac{x - 2}{\sqrt{x^2 - 2} - \sqrt{2}}$
- $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x^2 + 2x}$
- $\lim_{x \rightarrow 2} \frac{x - \sqrt{x+2}}{\sqrt{4x+1} - 3}$
- $\lim_{x \rightarrow a} \frac{x\sqrt{x} - a\sqrt{a}}{\sqrt{x} - \sqrt{a}}$
- $\lim_{x \rightarrow 0} \frac{1 - \sqrt[3]{1-x}}{3x}$

$$\begin{array}{lll}
19. \lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x^2} - 1}{x^2} & 20. \lim_{x \rightarrow 0} \frac{\sqrt{x+1} + \sqrt{x+4}}{x} & 21. \lim_{x \rightarrow 0} \frac{1}{(2+x)^3} - \frac{1}{8} \\
23. \lim_{x \rightarrow 0} \frac{a - \sqrt{a^2 - x^2}}{x^2} & 24. \lim_{x \rightarrow -2} \frac{x^3 + 8}{1 + \sqrt[3]{4x+7}} & 25. \lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x} - \sqrt[3]{1-x}}{x} \\
27. \lim_{x \rightarrow 8} \frac{\sqrt[3]{x} - 2}{\sqrt[3]{x+19} - 3} & 28. \lim_{x \rightarrow 1} \left(\frac{2}{x-1} - \frac{1}{\sqrt{x}-1} \right) & 29. \lim_{x \rightarrow 2} \left(\frac{2}{2-x} - \frac{1}{\sqrt{x}-2} \right) \\
31. \lim_{x \rightarrow 9} \frac{3x^3 - 81\sqrt{x}}{x - 3\sqrt{x}} & 32. \lim_{x \rightarrow 1} \frac{2 - 2\sqrt{x}}{x^2 - \sqrt{x}} & 33. \lim_{x \rightarrow \sqrt{2}} \frac{x^4 + 2x^2 - 8}{x^2 - 2} \\
35. \lim_{x \rightarrow 1} \frac{\sqrt[3]{x^2} - 2\sqrt[3]{x} + 1}{(x-1)^2} & 36. \lim_{x \rightarrow \sqrt{3}} \frac{x^2 - 3}{x^4 + x^2 + 1} & 37. \lim_{x \rightarrow 0} \frac{\sqrt[3]{x-1} + \sqrt{2x+1}}{x^2 - x} \\
38. \lim_{x \rightarrow 1} \frac{x^{200} - 1}{\sqrt[100]{x} - 1} & 39. \lim_{x \rightarrow 1} \frac{\sqrt[200]{x} - 1}{\sqrt[600]{x} - 1} & 40. \lim_{x \rightarrow 1} \frac{\sqrt[5]{2x^2-1} - 1}{x-1} \\
42. \lim_{x \rightarrow 2} \frac{\sqrt{x+7} + \sqrt{x+2} - 5}{x^2 - 4} & 43. \lim_{x \rightarrow 3} \frac{4x - \sqrt{x+1} - 10}{x^2 - 3x} & 44. \lim_{x \rightarrow 2} \frac{\sqrt{2x^2+1} - \sqrt{x+2} - 1}{x-2} \\
46. \lim_{x \rightarrow 1} \frac{(3x^3 - 6x^2 + 5x - 2)^{2000}}{(2x^2 - 3x + 1)^{2000}} & 47. \lim_{x \rightarrow 2} \frac{(x^3 - 12x + 16)^{500}}{(x^2 - x - 2)^{1000}} & 48. \lim_{x \rightarrow 1} \left(\frac{2+x}{2-x} \right)^{\frac{\sqrt{7+2x}-3}{x^2-1}} \\
& & 49. \lim_{x \rightarrow 2} \frac{x^8 + x - 258}{2x^6 - 2x^2 - 120}
\end{array}$$

VI. គណនាលីមីតខាងក្រោម

$$\begin{array}{lll}
1. \lim_{x \rightarrow +\infty} \frac{3x^2 + x + 4}{x^2 + 1} & 2. \lim_{x \rightarrow \pm\infty} \frac{4x^2 - 6x - 3}{x^2 - x - 6} & 3. \lim_{x \rightarrow +\infty} \frac{7x^2 - x + 8}{3x^2 + 2x} \\
5. \lim_{x \rightarrow +\infty} \frac{\sqrt{4x^2 + 5x} - 7}{2x - 3} & 6. \lim_{x \rightarrow \pm\infty} \frac{\sqrt{4x^2 - x} - 6}{2x + 1} & 7. \lim_{x \rightarrow +\infty} \frac{x+3}{\sqrt{25x^2 - 3x} + 1} \\
9. \lim_{x \rightarrow \pm\infty} \frac{\sqrt{9x^2 + 2x} - 1}{x+1} & 10. \lim_{x \rightarrow \pm\infty} \frac{3x-1}{\sqrt{x^2 - 2x} + 7} & 11. \lim_{x \rightarrow \pm\infty} \frac{(4x-1)(3-x)}{(2x-1)^2} \\
13. \lim_{x \rightarrow +\infty} \frac{x^2 + \sqrt[3]{x^2+2}}{x^3 - \sqrt{x+6}} & 14. \lim_{x \rightarrow +\infty} \frac{\sqrt{x^2+1} + \sqrt{x}}{\sqrt[4]{x^3+x} - x^4} & 15. \lim_{x \rightarrow +\infty} \frac{x^2 + \sqrt[3]{3x^2-2}}{x - \sqrt{x+6}} \\
17. \lim_{x \rightarrow \infty} \frac{2x^2 + 3}{x^3 - 2x + 1} & 18. \lim_{x \rightarrow \infty} \frac{(x-1)(2x+3)(2-x)}{(x^2+1)(2x+1)} & 19. \lim_{x \rightarrow \pm\infty} \frac{x^5 + 2x^2 + 1}{x^3 + 1} \\
21. \lim_{x \rightarrow \infty} \frac{(x+1)^{10} + (x+2)^{10} + \dots + (x+100)^{10}}{x^{10} - 10^{10}} & 22. \lim_{x \rightarrow \infty} \frac{(x-1)(2x+3)(2-x)}{(x^n+1)(2x+1)} & 23. \lim_{x \rightarrow +\infty} \frac{\sqrt{x^2+1} + \sqrt{x}}{\sqrt[4]{x^3+x} - x^4} \\
24. \lim_{x \rightarrow \infty} \frac{x^5 + (x+1)^5 + (x+2)^5 + \dots + (x+2026)^5}{x^5 - 200} & 25. \lim_{n \rightarrow +\infty} \frac{1+2+3+\dots+n}{n^2} & 26. \lim_{n \rightarrow +\infty} \frac{1^2 + 2^2 + 3^2 + \dots + n^2}{n^3}
\end{array}$$

V. គណនាលីមីតខាងក្រោម

$$\begin{array}{lll}
1. \lim_{x \rightarrow \infty} (-2x^2 + 5x - 34) & 2. \lim_{x \rightarrow +\infty} (7 + 2x^2 - 3x^3 - x^4) & 3. \lim_{x \rightarrow +\infty} (\sqrt{x^2 - 5x + 1} - \sqrt{x^2 + 3x - 3})
\end{array}$$

$$4. \lim_{x \rightarrow -\infty} (\sqrt{x^2 + 8x - 1} - \sqrt{x^2 - 3})$$

$$5. \lim_{x \rightarrow +\infty} (\sqrt{4x^2 + 3x - 1} - 2x)$$

$$6. \lim_{x \rightarrow -\infty} (\sqrt{4x^2 + 8x - 1} - \sqrt{x^2 - 3})$$

$$7. \lim_{x \rightarrow +\infty} (\sqrt{x + \sqrt{x}} - \sqrt{x})$$

$$8. \lim_{x \rightarrow +\infty} (\sqrt{x^2 - 3x} - x + 2)$$

$$9. \lim_{x \rightarrow +\infty} (\sqrt{4x^2 + 6x - 1} - 2x + 3)$$

$$10. \lim_{x \rightarrow +\infty} (\sqrt{4x^2 + 3x - 1} - 2x)$$

$$11. \lim_{x \rightarrow -\infty} (\sqrt{4x^2 + 10x - 1} + 2x + 5)$$

$$12. \lim_{x \rightarrow +\infty} (\sqrt{4x^2 - 3x - 1} - \sqrt{9x^2 + x})$$

VI. រកចំនួនពិត a និង b ដើម្បីឲ្យ

$$1. \lim_{x \rightarrow -2} \frac{x^2 + ax - 6}{2x^2 + 3x - 2} = b$$

$$2. \lim_{x \rightarrow 1} \frac{x^2 + ax + b}{x - 1} = 5$$

$$3. \lim_{x \rightarrow 2} \frac{x^2 - ax + 8}{x^2 - (b + 2)x + 2b} = \frac{1}{5}$$

$$4. \lim_{x \rightarrow 1} \frac{\sqrt{2ax} - 5}{x^2 - 1} = b$$

VII. គណនាលីមីតខាងក្រោម

$$1. \lim_{x \rightarrow 0} \frac{\sin 3x}{x}$$

$$2. \lim_{x \rightarrow 0} \frac{\sin x}{5x}$$

$$3. \lim_{x \rightarrow 0} \frac{\sin 10x}{x}$$

$$4. \lim_{x \rightarrow 0} \frac{\sin^2 2x}{x^2}$$

$$5. \lim_{x \rightarrow 0} \frac{\sin 2015x}{x}$$

$$6. \lim_{x \rightarrow 0} \frac{\sin 2x}{\sin 12x}$$

$$7. \lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx}$$

$$8. \lim_{x \rightarrow 0} \frac{\tan kx}{x}$$

$$9. \lim_{x \rightarrow 0} \frac{x \tan x}{\sin^2 x}$$

$$10. \lim_{x \rightarrow 0} \frac{5 \sin^2 \frac{1}{5} x}{x^2}$$

$$11. \lim_{x \rightarrow 0} \frac{1 - \cos x}{2x}$$

$$12. \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x \sin 5x}$$

$$13. \lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{\sin 5x}$$

$$14. \lim_{x \rightarrow 0} \frac{\cos 6x - \cos x}{\sin^2 x}$$

$$15. \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$$

$$16. \lim_{x \rightarrow \frac{\pi}{4}} \frac{\cos x - \sin x}{\cos 2x}$$

$$17. \lim_{x \rightarrow 0} \frac{1 - \cos x^3}{\sin^2 x}$$

$$18. \lim_{x \rightarrow 0} \frac{\sin^3 2x}{x^2 \sin 6x}$$

$$19. \lim_{x \rightarrow 0} x \cot x$$

$$20. \lim_{x \rightarrow 0} \frac{x \tan \frac{x}{2}}{1 - \cos 2x}$$

$$21. \lim_{x \rightarrow 0} \frac{x}{\sin(\sin 3x)}$$

$$22. \lim_{x \rightarrow 0} \frac{\sin(\tan^2 x)}{x^2}$$

$$23. \lim_{x \rightarrow 0} \frac{\sqrt{x+2} - \sqrt{2}}{\sin 3x}$$

$$24. \lim_{x \rightarrow 0} \frac{-6 \sin 10x}{\sqrt{x+9} - 3}$$

$$25. \lim_{x \rightarrow 0} \frac{\sqrt{2x+1} - \sqrt{x+1}}{\sin x}$$

$$26. \lim_{x \rightarrow 0} \frac{\sqrt{1+\sin x} - \sqrt{1-\sin x}}{\tan x}$$

$$27. \lim_{x \rightarrow 0} \frac{\sqrt{2x^2+9} - 3}{3 \sin^2 x}$$

$$28. \lim_{x \rightarrow 0} \frac{\sin 2x}{\sqrt{x+3} - \sqrt{2x+3}}$$

$$29. \lim_{x \rightarrow 0} \frac{1 - \sqrt{\cos x}}{\sin^2 x}$$

$$30. \lim_{x \rightarrow 0} \frac{\sqrt{5} + \sqrt{2+3 \cos x}}{x \sin x}$$

$$31. \lim_{x \rightarrow 0} \frac{1 - \cos 5x}{1 - \sqrt{x^2 + 1}}$$

$$32. \lim_{x \rightarrow 0} \frac{x \sin 4x}{\sqrt{2x^2+2} - \sqrt{2}}$$

$$33. \lim_{x \rightarrow 0} \frac{4 \sin 4x}{3x - \sin 2x}$$

$$34. \lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x}{1 - \sin x - \cos x}$$

$$35. \lim_{x \rightarrow 0} \frac{\sin x \sin 2x \dots \sin nx}{x^n}$$

$$36. \lim_{x \rightarrow 0} \frac{\sin x + \sin 2x + \dots \sin nx}{x}$$

$$37. \lim_{x \rightarrow 0} \frac{\sin x + \sin 3x + \sin 5x}{\sin 6x + \sin 8x + \sin x}$$

$$38. \lim_{x \rightarrow 0} \frac{\sin(202\pi + 2x)}{\sin(203\pi + 3x)}$$

$$39. \lim_{x \rightarrow \pi} \frac{\sin(x - \pi)}{x - \pi}$$

$$40. \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin\left(x - \frac{\pi}{2}\right)}{x - \frac{\pi}{2}}$$

$$41. \lim_{x \rightarrow -\pi} \frac{1 - \cos x}{x + \pi}$$

$$42. \lim_{x \rightarrow \pi} \frac{\cos x + 1}{x - \pi}$$

$$43. \lim_{x \rightarrow -\frac{\pi}{2}} \frac{\cos x}{x + \frac{\pi}{2}}$$

$$44. \lim_{x \rightarrow -2} \frac{1 - \cos(x+2)}{x+2}$$

$$45. \lim_{x \rightarrow 1} \frac{\sin \pi x}{\sin 3x}$$

$$46. \lim_{x \rightarrow 1} \frac{\cos \frac{\pi x}{2}}{\sin \pi x}$$

$$\begin{aligned}
47. \lim_{x \rightarrow \pi} \frac{1 + \cos x}{(x - \pi)^2} & \quad 48. \lim_{x \rightarrow \frac{\pi}{3}} \frac{\sin x - \sqrt{3} \cos x}{2(\pi - 3x)} & 49. \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x - \cos x}{4x - \pi} & 50. \lim_{x \rightarrow 0} \frac{2x - \sin x}{\sqrt{1 - \cos x}} \\
51. \lim_{x \rightarrow \frac{\pi}{6}} \frac{2 \sin^2 x - 3 \sin x + 1}{4 \sin^2 x - 1} & 52. \lim_{x \rightarrow \frac{\pi}{2}} \frac{(1 - \cos x)^2}{\tan^3 x - \sin^3 x} & 53. \lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin x}{\left(\frac{\pi}{2} - x\right)^2} & 54. \lim_{x \rightarrow \pm\infty} x \sin \frac{1}{x} \\
55. \lim_{x \rightarrow +\infty} x^2 \left(1 - \cos \frac{1}{x}\right) & 56. \lim_{x \rightarrow 0} \frac{1 - \cos x \cos 2x}{x^2} & 57. \lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{1 + x \sin x} - \cos x} & 58. \lim_{x \rightarrow 0} \frac{x(a-b)}{\sin ax - \sin bx}
\end{aligned}$$

VIII. គណនាលីមីតខាងក្រោម

$$\begin{aligned}
1. \lim_{x \rightarrow 1} \frac{4 \sin(x-1)}{1-x} & \quad 2. \lim_{x \rightarrow 1} \frac{-3 \sin(x-1)}{1-x^2} & 3. \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\pi}{2} - x\right) \tan x & 4. \lim_{x \rightarrow 0} \left(\frac{1}{x^2} \left(\frac{2}{\cos x} + \cos x - 3\right)\right) \\
5. \lim_{x \rightarrow 0} \frac{2x - \sin 2x}{x^3} & 6. \lim_{x \rightarrow 0} \frac{2x - \sin 2x}{x - \sin x} & 7. \lim_{x \rightarrow 0} \frac{3x - \sin 3x}{5x - \sin 5x} & 8. \lim_{x \rightarrow 0} \frac{3 \tan x - \tan 3x}{\tan 2x - \sin 2x} \\
9. \lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x} & 10. \lim_{x \rightarrow 0} \frac{e^{5x} - 1}{4x} & 11. \lim_{x \rightarrow 0} \frac{\sin x + e^x - 1}{x^2 + x} & 12. \lim_{x \rightarrow 0} \frac{\sin^2 x + e^{x^2} - 1}{x^3 + 6x^2} \\
13. \lim_{x \rightarrow 0} \frac{e^{\sin x} - 1}{x} & 14. \lim_{x \rightarrow 0} \frac{e^{x^2} - 1}{1 - \cos x} & 15. \lim_{x \rightarrow 0} \frac{e^{2x} - e^{-2x}}{x} & 16. \lim_{x \rightarrow 0} \frac{e^{\sin x} - e^{-2x}}{x} \\
17. \lim_{x \rightarrow 0} \left(\frac{e^x}{x} - \frac{1}{x} + \frac{\sin 2x}{x}\right) & 18. \lim_{x \rightarrow \frac{4}{4}} \frac{\cos x - \sin x}{\cos 2x} & 19. \lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x}\right)^{3x} & 20. \lim_{x \rightarrow +\infty} \left(\frac{x+1}{x}\right)^x \\
21. \lim_{x \rightarrow 0} (1 + 2x)^{\frac{3}{x}} & 22. \lim_{x \rightarrow 0} (1 + 4x)^{\frac{3}{x}} & 23. \lim_{x \rightarrow 0} (1 + \sin 5x)^{\frac{2}{x}} & 24. \lim_{x \rightarrow 0} \left(\frac{4x+1}{x+1}\right)^{\frac{4}{x}} \\
25. \lim_{x \rightarrow 0} \left(\frac{3x+2}{2x+2}\right)^{\frac{5}{x}} & 26. \lim_{x \rightarrow 0} \left(\frac{2x^2+3x+1}{2x^2+1}\right)^{\frac{3}{2x}} & 27. \lim_{x \rightarrow 0} \frac{xe^{x+1} - ex - 1 + \cos 2x}{1 - \cos x} & \\
28. \lim_{x \rightarrow 0} \frac{e^{x^2} - \cos 4x}{x^2} & 29. \lim_{x \rightarrow 0} \frac{\ln(1 + \tan 2x)}{x} & 30. \lim_{x \rightarrow 0} \frac{\ln(\cos 2x)}{x^2} & 31. \lim_{x \rightarrow 0} \frac{\ln(1 + 4x - x^2)}{x} \\
32. \lim_{x \rightarrow +\infty} [\ln(e^{x+1} + 1) - x] & 33. \lim_{x \rightarrow +\infty} x[\ln(x+1) - \ln x] & 34. \lim_{x \rightarrow \pm} (x^2 + xe^x) & 35. \lim_{x \rightarrow +\infty} (1-x)e^x \\
36. \lim_{x \rightarrow +\infty} (x+2)e^{-x} & 37. \lim_{x \rightarrow +\infty} \frac{e^x - x}{2e^x + 1} & 38. \lim_{x \rightarrow 0} \ln\left(\frac{x}{x+1}\right) & 39. \lim_{x \rightarrow \pm\infty} x \ln(x^2 + 1) \\
40. \lim_{x \rightarrow 0} \frac{\ln(1+2x)}{x} & 41. \lim_{x \rightarrow 0} \frac{\ln(1+3x)}{x^2 + 4x} & 42. \lim_{x \rightarrow 0} \frac{\ln(1-7x)}{x} & 43. \lim_{x \rightarrow 0} \frac{\ln(1+2x+x^2)}{4x+2x^3} \\
44. \lim_{x \rightarrow 0} \frac{\ln(3-2e^x)}{x} & 45. \lim_{x \rightarrow 0} \frac{\ln[(1+mx)(1+nx)]}{x} & 46. \lim_{x \rightarrow 0} \frac{\ln(1+2 \sin 3x)}{\sin 2x} & 47. \lim_{x \rightarrow 0} \frac{\ln(2 - e^{\sin^2 x})}{x \tan x} \\
48. \lim_{x \rightarrow 0} \frac{\ln(\cos x) + \ln(\cos 2x)}{x^2} & 49. \lim_{x \rightarrow 0} \frac{\ln x + \sin 3x}{x} & 50. \lim_{x \rightarrow 3} \frac{x^3 - 3^x}{x - 3} & 51. \lim_{x \rightarrow 0} \frac{4^x - 2^x}{\sin 2x} \\
52. \lim_{x \rightarrow 0} \frac{\ln x + \sin 3x}{x} & 53. \lim_{x \rightarrow 0} \frac{(3e^x - 3)(1 - \cos 2x)}{x^2 \sin 2x} & 54. \lim_{x \rightarrow 0} \frac{4^x - 2^x}{\sin 2x} & 55. \lim_{x \rightarrow e} \frac{2 \ln^2 x + \ln x - 3}{2x - 2e} \\
56. \lim_{x \rightarrow e} \frac{3 \ln^3 x - 3 \ln^2 x}{x^2 - e^2} & 57. \lim_{x \rightarrow 0} \frac{2 \cos^2 x + 2 \cos x - 4}{x^2} & 58. \lim_{x \rightarrow \frac{\pi}{6}} \frac{4 \sin^2 x + 2 \sin x - 2}{2 \sin x - 1} & 59. \lim_{x \rightarrow 0} \frac{4^x \sin x - 2^x \sin x}{x^2} \\
60. \lim_{x \rightarrow e} \frac{3 \ln^2 x - \ln x - 2}{x - e} & 61. \lim_{x \rightarrow 0^+} \frac{4 \ln^2 x + 3 \ln x + 100}{\ln^2 x - 2} & 62. \lim_{x \rightarrow +\infty} \frac{\sqrt[3]{8x^3} - \sqrt[3]{x} + \sqrt{x} - 1}{\sqrt[3]{x^3} - \sqrt{x} - \sqrt[4]{x^2} - x} & 63. \lim_{x \rightarrow 0} \frac{\sin(\sin x)}{\sin(\sin(\sin(\sin x)))}
\end{aligned}$$

$$64. \lim_{x \rightarrow +\infty} \frac{\sqrt{x} + \sqrt[3]{x} + \sqrt[4]{x}}{\sqrt{x} + \sqrt{4x} - 1}$$

$$67. \lim_{x \rightarrow -2} \frac{x^4 - 5x^3 + 2x^2 - 3x - 70}{x^2 + x - 2}$$

$$70. \lim_{x \rightarrow 2} \frac{x^2 - 4 + \sin \pi x}{x - 2}$$

$$73. \lim_{x \rightarrow 0} \frac{e^{3x} + \sin 4x - 1}{x}$$

$$76. \lim_{x \rightarrow 0} \frac{e^{1-\sqrt{\cos x}} - 1}{3 \sin x \tan 2x}$$

$$79. \lim_{x \rightarrow \infty} \frac{2x+3}{\sqrt[3]{x^3+4x} + \sqrt[3]{8x^3+1}}$$

$$82. \lim_{x \rightarrow e} \frac{3 \ln^2 x - \ln x - 2}{x - e}$$

$$85. \lim_{n \rightarrow +\infty} \frac{1^2 + 2^2 + 3^2 + \dots + n^2}{n(n^2 + 5)}$$

$$88. \lim_{n \rightarrow +\infty} \frac{1+2+3+\dots+n}{\sqrt{9n^4+1}}$$

$$65. \lim_{n \rightarrow +\infty} \frac{(n+2)! + n!}{(n+2)! + (n+1)!}$$

$$68. \lim_{x \rightarrow 0} \frac{1 - \cos(1 - \cos x)}{x^4}$$

$$71. \lim_{x \rightarrow 1} \frac{x^3 - 1 + \tan \pi x}{1 - x^2}$$

$$74. \lim_{x \rightarrow 0} \frac{(e^{3x+2} - e^2) \sin(\pi x)}{x^2}$$

$$77. \lim_{x \rightarrow 0} \frac{3\sqrt{\cos^3 x} e^{2 \cos 8x-1} - 3e}{5e^{2x} - 10x + 5}$$

$$80. \lim_{x \rightarrow \infty} \frac{\sqrt[3]{27x^3+4} + 3\sqrt[3]{8x^3+1}}{x + \sqrt[5]{32x^5+1}}$$

$$83. \lim_{x \rightarrow e} \frac{2 \ln^2 x + \ln x - 3}{2x - 2e}$$

$$86. \lim_{n \rightarrow +\infty} \frac{1^2 + 2^2 + 3^2 + \dots + n^2}{n(1+2+3+\dots+n)}$$

$$89. \lim_{x \rightarrow 1} \frac{\sqrt{x^3-1} + x^2 - \sqrt{x-1} - 1}{2\sqrt{x-1} + \sqrt{x^2-1}}$$

$$66. \lim_{n \rightarrow +\infty} \frac{2+4+6+\dots+2n}{n^2-2n}$$

$$69. \lim_{x \rightarrow 1} \frac{\sin(x^2-1)}{x\sqrt{x}-1}$$

$$72. \lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{\sin 5x}$$

$$75. \lim_{x \rightarrow 0} \frac{1 - 2e^{3 \sin 7x} - \cos x}{2x + x^2}$$

$$78. \lim_{x \rightarrow \infty} \left[\left(x + \frac{2}{x} \right)^2 - \left(x - \frac{2}{x} \right)^2 \right]$$

$$81. \lim_{x \rightarrow 3} \frac{x^3 - 3^x}{x - 3}$$

$$84. \lim_{x \rightarrow e} \frac{3 \ln^3 x - 3 \ln^2 x}{x^2 - e^2}$$

$$87. \lim_{x \rightarrow +\infty} [\sqrt{(x+a)(x+b)} - 2x]$$

$$90. \lim_{x \rightarrow 1} \frac{\sqrt{x+1} + \sqrt{x^2-1} - \sqrt{x^3+1}}{\sqrt{x-1} + \sqrt{x^3+1} - \sqrt{x^4+1}}$$

91. គណនា $S_n = \frac{1}{1 \cdot 4 \cdot 7} + \frac{1}{4 \cdot 7 \cdot 10} + \dots + \frac{1}{(3n-2)(3n+1)(3n+4)}$ ។ ទាញរក $\lim_{n \rightarrow +\infty} \sum_{k=1}^n \frac{1}{(3k-2)(3k+1)(3k+4)}$

92. គេឲ្យអនុគមន៍ $f(x) = \frac{ax^3 + bx^2 + 4}{x-2}$ ។ កំណត់ចំនួនពិត a និង b ដើម្បីអោយ $\lim_{x \rightarrow 2} f(x) = 20$ ។

93. កំណត់តម្លៃ a ដែលបំពេញលក្ខខណ្ឌ $\lim_{x \rightarrow 0} \frac{1+ax-\sqrt{1+x}}{x} = \frac{1}{8}$ ។

94. កំណត់អនុគមន៍ដឺក្រេទី២ $y = f(x)$ ដែលបំពេញលក្ខខណ្ឌលីមីតទាំងពីរខាងក្រោម

ក. $\lim_{x \rightarrow +\infty} \frac{f(x)}{x^2-1}$

ខ. $\lim_{x \rightarrow 1} \frac{f(x)}{x^2-1} = -1$

95. កំណត់តម្លៃ $\lim_{x \rightarrow a} \frac{x^2 - ax + 1}{x-1}$ ជាចំនួនថេរ។ គណនាលីមីតនេះ។

96. កំណត់តម្លៃ a និង b ដើម្បីបំពេញលក្ខខណ្ឌ $\lim_{x \rightarrow -2} \frac{x^2 + ax - 6}{2x^2 + 3x - 2} = b$

97. កំណត់ចំនួនពិត a ដើម្បីអោយលីមីតខាងក្រោមជាលីមីតនៃចំនួនថេរ ហើយកំណត់លីមីតនេះ

ក. $\lim_{x \rightarrow 1} \frac{\sqrt{x+3} - a}{x-1}$

ខ. $\lim_{x \rightarrow 0} \frac{\sqrt{1+3x} + a}{x}$

គ. $\lim_{x \rightarrow 2} \frac{\sqrt{x+a} - 1}{x-2}$

ឃ. $\lim_{x \rightarrow -1} \frac{\sqrt{x^2+ax} - 1}{x^2-1}$

98. គេមាន $f(x) = \frac{x}{\sqrt{1+x^2}}$ ។ គណនា $f_n(x) = f(f \dots f(x) \dots)$ រួចគណនា $\lim_{n \rightarrow +\infty} f_n(x)$ ។